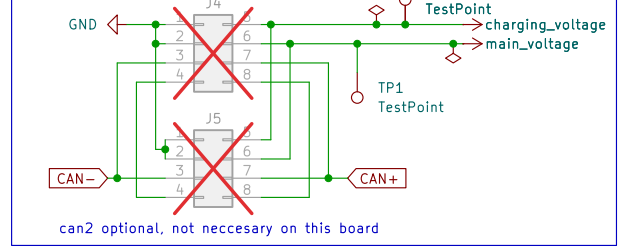


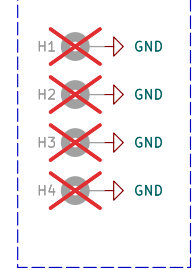
Power Board

Connectors

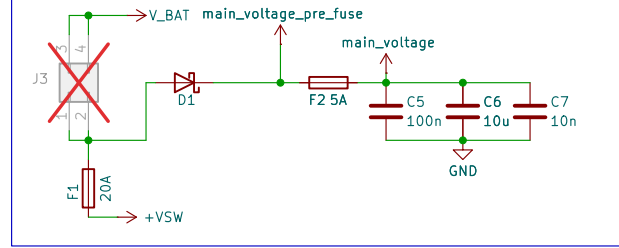
Vehicle BUS



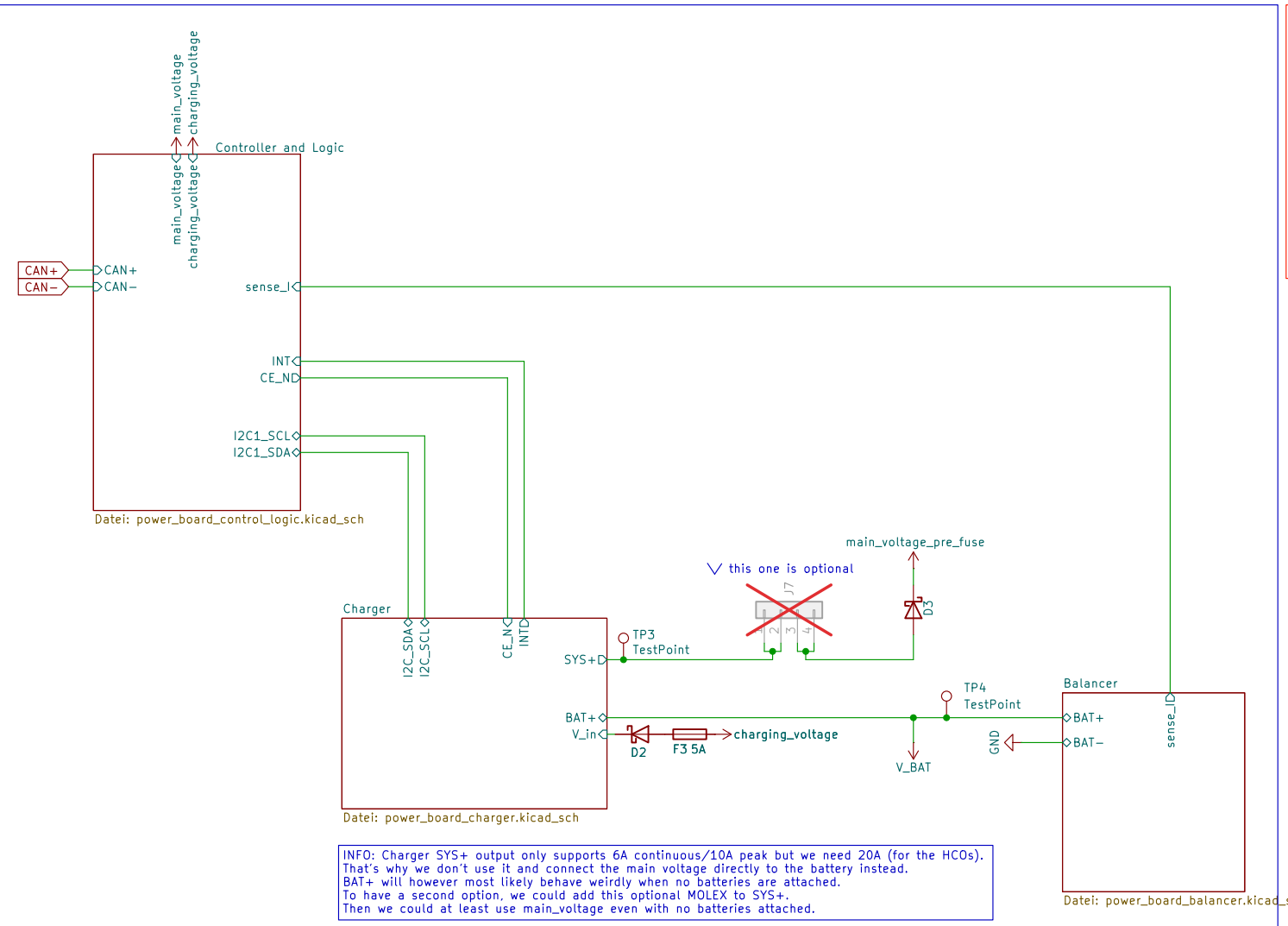
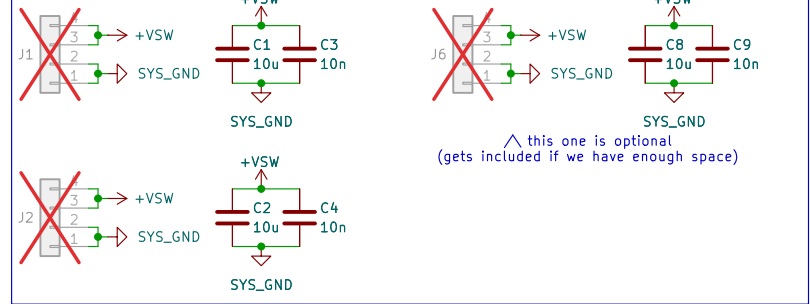
mounting



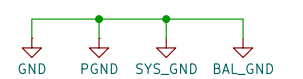
Main Switch



Power Connectors



GND connections



the pcb shall be routed in a way which ensures that each distinct GND has a dedicated current loop which does not cross with other current loops. This is supposed to separate different current loops from eachother this should decrease crosstalk problems.

TUDSaT

Sheet: /
File: power_board.kicad_sch

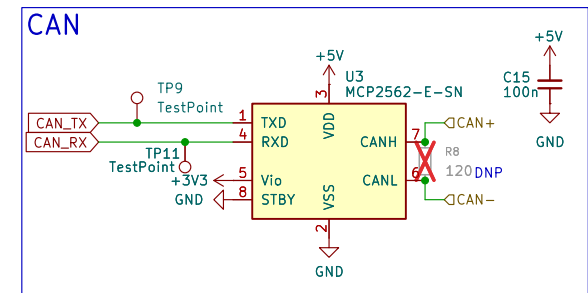
Title: Power Board

Size: A3 Date: 2026-07-01

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Rev: 4
Id: 1/4

programming adapters



Regulators

redundant but does not hurt

The diagram illustrates two voltage regulators. The top regulator, U4 (L7805), is a 5V regulator. Its input (IN, pin 1) is connected to a diode OR gate formed by D4 and D5, which combines the main_voltage and charging_voltage signals. A 330nF capacitor (C13) is connected between the input and ground. The output (OUT, pin 3) is connected to a +5V output line, with a 100uF capacitor (C14) connected between the output and ground. The bottom regulator, U1 (AZ1117I-3.3), is a 3.3V regulator. Its input (VI, pin 3) is connected to a +5V input line (labeled TP5 TestPoint). A 10uF capacitor (C11) is connected between the input and ground. The output (VO, pin 2) is connected to a +3V3 output line (labeled TP12 TestPoint). A 22uF capacitor (C12) is connected between the output and ground. Both regulators have their ground pins (pin 2 for U4, pin 1 for U1) connected to a common GND.

Sensing

24V
charging_voltage

R10
68k

sense_V_charge

R11
10k

GND

12.6V
main_voltage

R12
30k

sense_V_main

R13
10k

GND

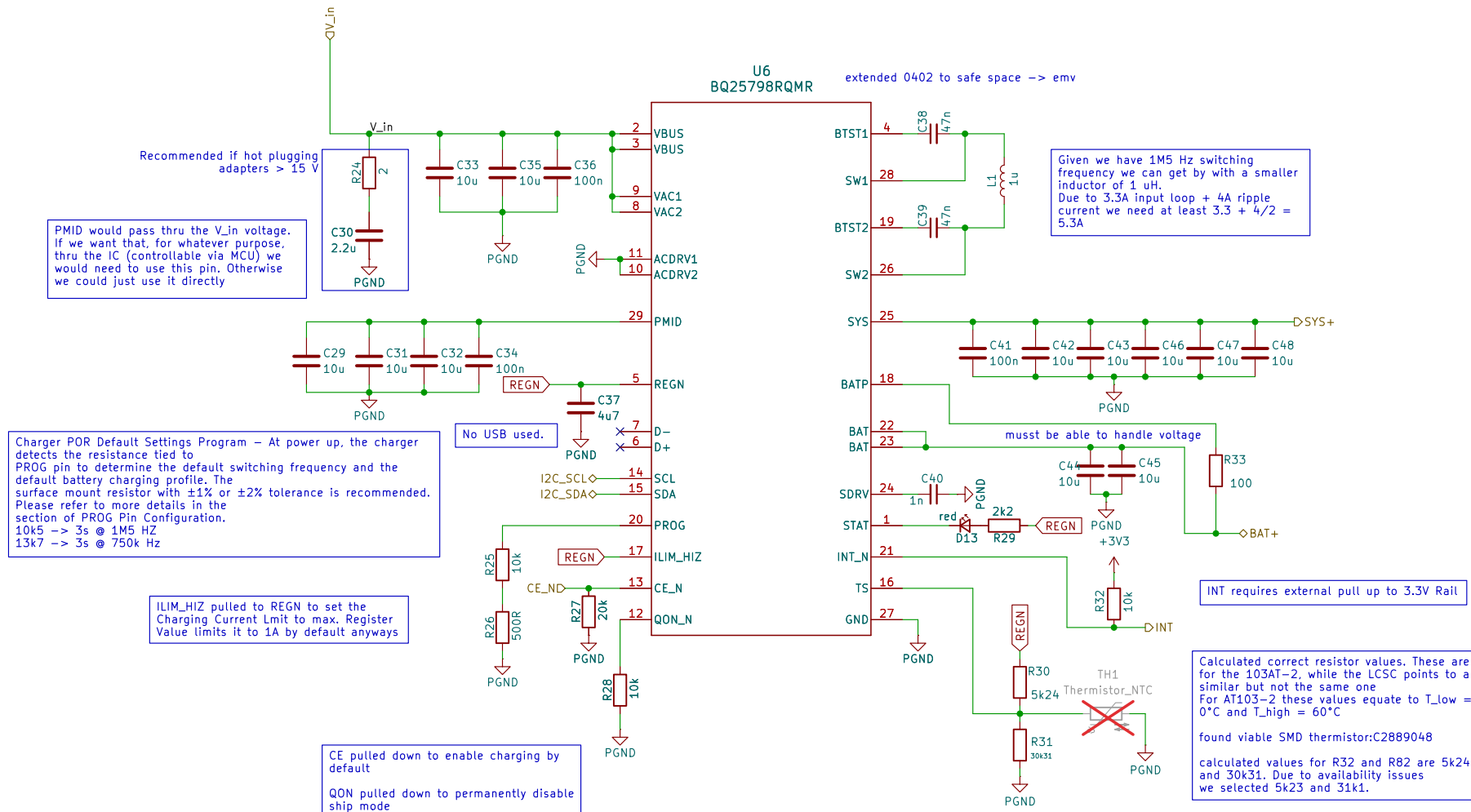
$$V_{CH_s}/24V = R19/(R18+R19)$$
$$V_{CH_s_max} = 3v3; R19 = 10k$$
$$\rightarrow R18 = 68k$$
$$V_{SNS}/12.6V = R21/(R20+R21)$$
$$V_{SNS_max} = 3v3; R21 = 10k$$
$$\rightarrow R20 = 30k$$

LEDs

Clock

[illegible]

Buck-Boost-Converter



Maybe a useful link to a schematics checklist. But for another charger.
https://e2e.ti.com/cfs-file/_key/communityserver-discussions-components-files/196/BQ25750_5F00_Schematic_5F00_And_5F00_Layout_5F00_Checklist.pdf

TUDSaT

Sheet: /Charger/
File: power_board_charger.kicad_sch

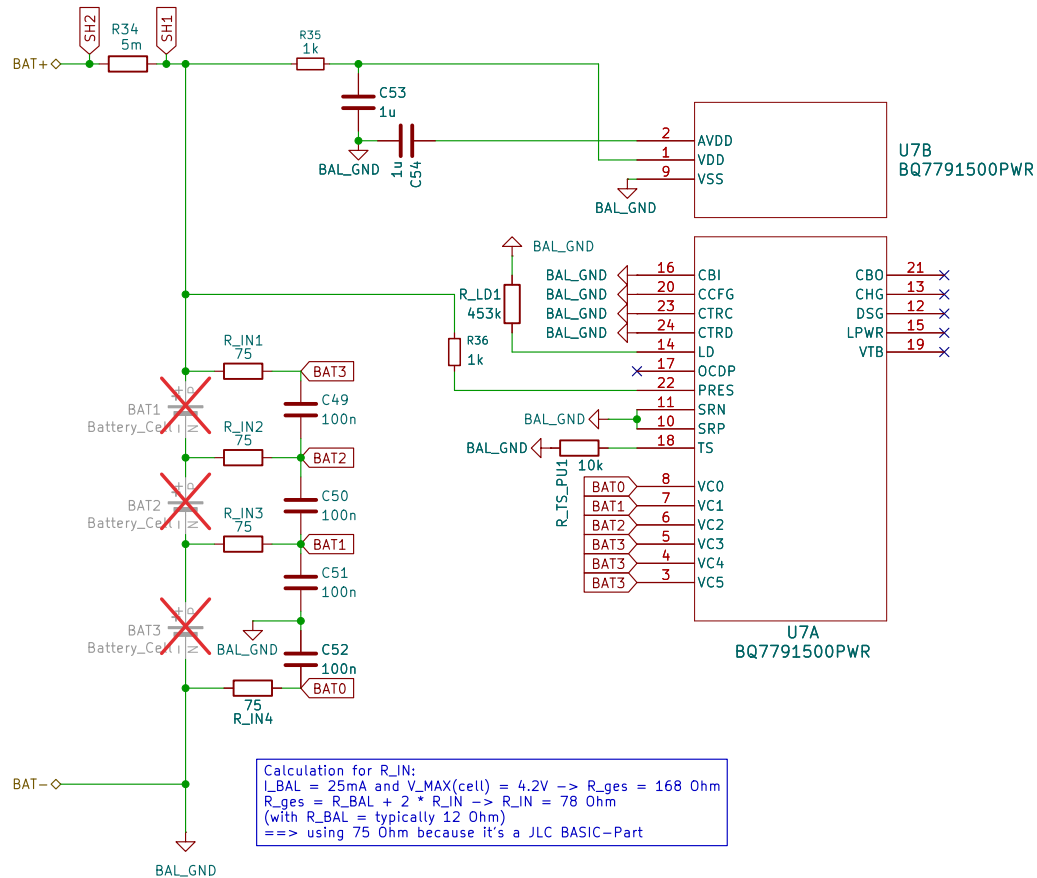
Title: Power Board (Charger)

Size: A4 Date: 2026-07-01

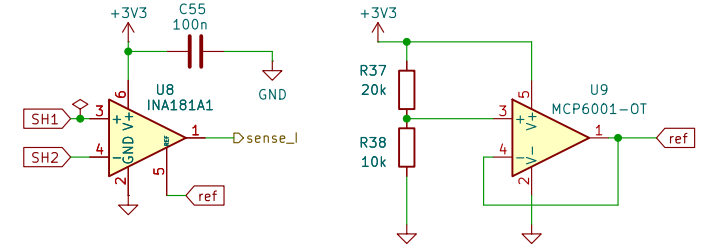
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Rev: 4

Id: 4/4



Sensing



For best performance, keep the source impedance to the REF pin less than 5 Ω .

Regarding the thermistor:

The tipping-points of our balancer are – BY DESIGN – 45°C for OTC and 65°C for OTD as well as 0°C UTC and –10°C UTD (the latter 2 are not very relevant for us). In order to trigger those, for V_{TS} (i.e. the voltage across the thermistor) must hold: $V_{TS} = 0.2056 * V_{TB}$ (for OTD) and $V_{TS} = 0.3294 * V_{TB}$ (for OTC). Then R_{TS} can be calculated as $R_{TS} = R_{TS_PU} * V_{TS} / (V_{TB} - V_{TS})$ with fixed R_{TS} . For the upper mentioned voltage ratios we get $R_{TS_PU} = 3.864 * R_{TS}$ (for OTD) and $R_{TS_PU} = 2.036 * R_{TS}$ (for OTC). For fixed $R_{TS_PU} = 10k$ we can conclude that the thermistor must satisfy:

$$R_{TH@45^{\circ}C} = 4,9k \text{ and } R_{TH@65^{\circ}C} = 2,6k$$

INFO: to circumvent temp. prot. we can simply connect TS to GND through a 10k resistor.

TUDSaT

Sheet: /Balancer/

File: power_board_balancer.kicad_sch

Title: Power Board (Balancer)

Size: A4

Date: 2026-07-01

Rev: 4

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Id: 6/4